CORE ENTERPRISE ERP • MANUFACTURING & QUALITY MANAGEMENT LICENSE (ERPMAN)

SAGE X3 V12 ENTERPRISE MANUFACTURING & QUALITY CONTROL (ERPMAN)

TECHNICAL OPERATIONS MANUAL & STANDARD OPERATING PROCEDURE

Standard Operating Procedure (SOP) and Technical Reference Guide for Production Engineering (BOM & Routings), Work Order Execution (`GESMFG`), Shop Floor Tracking (`GESMTK`), Quality Control & Statistical Sampling (`QLYCRD` / `STOQLYSMP`), Stock Quality Status Management (`Status Q/A/R`), Production Costing (`MFGCOST`), and Database Schema Architecture.

DOCUMENT CONTROL & TECHNICAL GOVERNANCE METADATA

Document Title:	**Sage X3 v12 Enterprise Manufacturing & Quality Control (ERPMAN) Technical Operations Manual**
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VERSION	EFFECTIVE DATE	SUMMARY OF REVISIONS	AUTHORIZING OFFICER
v1.0 Official	11 September 2026	Complete technical documentation of ERPMAN module, Quality Control (QC/QA) lifecycle, Database.sql schema, and shop floor execution.	Muhammad Iskandar (IT)

1.0 INTRODUCTION, ENTERPRISE ARCHITECTURE & ERPMAN SCOPE

The **Sage X3 v12 Manufacturing & Quality Control (`ERPMAN`)** module forms the industrial production core and compliance backbone of the BubblesO2 Enterprise Resource Planning (ERPFULL) ecosystem. It transforms raw materials procured via `ERPDIS` into certified natural mineral water while executing real-time Quality Control (QC/QA) inspections mandated by the Ministry of Health Malaysia (KKM), HACCP, and Halal standards.

1.1 Manual Shop Floor Practices vs Automated Sage X3 Manufacturing Engine

OPERATIONAL DIMENSION	LEGACY MANUAL LIMITATIONS	SAGE X3 AUTOMATED MANUFACTURING ENGINE
Production Planning	Manual whiteboards and paper job sheets. No visibility into warehouse raw material stock reservations.	**Work Order Scheduling (`GESMFG`):** Automated requirement planning, line scheduling, and component stock reservation.
Material Consumption	Post-shift physical counting. High rate of unaccounted material wastage and inaccurate standard costs.	**Real-Time Backflushing (`GESMTK`):** Direct deduction of preforms, caps, labels, and boxes based on actual finished output.
Quality Control (QC)	Disconnected paper test logs. High risk of uninspected or contaminated bottled water released to customers.	**Integrated Quality Inspection (`GESQCR` / `GESSMP`):** Automatic quarantine (**Status Q**), ISO sampling, and digital COA release.
WIP & Cost Variance	Cost variances discovered months later during annual audits. Inability to isolate line loss by work center.	**Automated WIP Settlement (`CSTMFG`):** Continuous tracking of material usage variances, labor hours, and machine overheads.

1.2 Comprehensive Navigation Map: Top 12 Manufacturing & QC Fast Codes

Plant supervisors and QA personnel navigate directly using these 6-character function codes in the Global Navigation Bar:

FAST CODE	MODULE DOMAIN	MENU NAVIGATION PATH	CORE BUSINESS PURPOSE
`GESBOD`	Engineering	*Manufacturing > Master data > Production BOMs*	Bill of Materials recipe management, component ratios, and scrap factors.
`GESROU`	Engineering	*Manufacturing > Master data > Routings*	Operational routing sequences, machine cycle rates, and quality sheet links.
`GESMWS`	Engineering	*Manufacturing > Master data > Work centers*	Configuration of plant machines (Blow, Fill, Label, Pack, Palletize) and capacity.
`GESMFG`	Work Orders	*Manufacturing > Planning > Work orders*	Work order generation, target scheduling, component reservation, and traveler ticket.
`GESMTK`	Shop Floor	*Manufacturing > Execution > Production tracking*	Recording good cartons, scrap rejects, machine hours, and material backflushing.
`GESQCR`	Quality Control	*Stock > Quality control > Quality control*	Technical inspection sheets, recording chemical/microbiological test readings.
`GESQST`	Quality Control	*Common data > Quality > Questions*	Definition of inspection questions, numeric thresholds (pH, TDS, torque), and limits.
`GESSMP`	Quality Control	*Stock > Quality control > Sampling*	Statistical sampling plans (ISO 2859 / AQL) for batch inspection.
`GESSCS`	Quality Control	*Stock > Internal movements > Stock change*	Formal quality status conversion (**Q → A** for accepted, **Q → R** for rejected).
`CONSSST`	Inventory	*Stock > Inquiries > Stock by site*	Multi-status inventory inquiry monitoring stock balances in Status A, Q, and R.
`CSTMFG`	Costing	*Costing > Production costs > Work order calculation*	Roll-up of actual vs standard production costs and WIP variance calculation.
`FUNMCLO`	Work Orders	*Manufacturing > Utilities > Close work orders*	Formal administrative and financial closure of completed production orders.

PLANT GOVERNANCE & SEPARATION OF DUTIES (SOD CONTROLS)

In compliance with food safety and financial audit standards, shop floor operators cannot modify BOM recipes or release inventory from Quarantine (Status Q). Quality Assurance (QA) officers cannot adjust physical warehouse stock without an authorized laboratory inspection record.

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2.0 ENGINEERING MASTER DATA: PRODUCTION BOMS & ROUTINGS

Production accuracy in Sage X3 relies on engineering master data: **Bill of Materials (`BOM`)** defining raw material formulations and **Manufacturing Routings (`ROUTING`)** governing machine sequences.

2.1 Bill of Materials (BOM) Hierarchy & Scrap Factors (GESBOD)

The master BOM defines the exact component ratios needed to pack 1 carton of mineral water: **Parent Product:** `FG-425X24` (BubblesO2 Natural Mineral Water 425ml × 24 Bottles) • **Alternative:** `1` (Standard Production)

COMPONENT CODE	COMPONENT DESCRIPTION	QTY / CTN	UOM	SCRAP (`SCA_0`)	OPERATIONAL POINT OF CONSUMPTION
`RM-PREF-425`	PET Bottle Preform 425ml	24.00	PCS	1.00%	Op 10: Rotary Blow Molding Machine
`RM-CAP-BLUE`	HDPE Bottle Screw Cap 28mm	24.00	PCS	0.50%	Op 20: Cleanroom Capping Station
`RM-LBL-425`	BOPP Printed Shrink Label 425ml	24.00	PCS	1.50%	Op 30: Steam Tunnel Labeling Machine
`RM-BOX-425`	Corrugated Printed Carton Box 425ml × 24	1.00	PCS	0.50%	Op 40: Automated Case Packer
`RAW-WATER`	Natural Mineral Borehole Water	10.20	LTR	2.00%	Op 20: Class 100 Cleanroom Filling Monoblock

2.2 Manufacturing Routings & Machine Work Centers (GESROU / GESMWS)

The routing establishes the five continuous operational work centers across Bottling Line 1 and Line 2:

OP #	WORK CENTER (`WST_0`)	OPERATION NAME	RATED SPEED	QC CONTROL SHEET	OPERATIONAL DESCRIPTION
Op 10	`WST-BLOW01`	Rotary Blow Molding	12,000 BPH	`QC-PREF`	Heating preforms to 110°C; high-pressure blowing of 425ml bottles.
Op 20	`WST-FILL01`	Cleanroom Fill & Cap	12,000 BPH	`QC-BOTT` (Critical)	Class 100 Cleanroom 0.2µ microfiltration, filling, and cap torque seal.
Op 30	`WST-LABL01`	Optical Sleeve Label	12,000 BPH	`QC-LBL`	Hot-melt BOPP sleeve application and steam tunnel shrinking.
Op 40	`WST-PACK01`	Automated Case Packer	500 CTN/Hour	`QC-PACK`	Robotic collating of 24 bottles into corrugated carton and taping.
Op 50	`WST-PALL01`	Palletizer & Wrapper	20 Pallets/Hour	`QC-PALL`	Robotic pallet stacking (72 cartons/pallet) and stretch film wrapping.

CRITICAL CONTROL POINT (CCP) LINKAGE IN ROUTING OPERATIONS

In compliance with HACCP standards, Operation 20 (Cleanroom Fill & Cap) is strictly tagged with Technical Sheet `QC-BOTT` in field `TECCRD_0` . Sage X3 automatically enforces quality inspection questions whenever tracking is reported on this work center.

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3.0 WORK ORDER GENERATION, SCHEDULING & MATERIAL ALLOCATION

Work Orders govern all shop floor production activities, linking sales demand or stock replenishment targets to machine scheduling and raw material reservations.

FLOWCHART 1: END-TO-END MANUFACTURING ORDER & ALLOCATION LIFECYCLE



3.1 Step-by-Step Configuration & Mandatory Screen Fields (GESMFG)

STEP #	SCREEN FIELD NAME	DATABASE FIELD (MFGHEAD)	OPERATIONAL GUIDANCE & FIELD VALUES
1	Manufacturing Site	`MFGFCY_0`	Select `BOMSB` (BubblesO2 Manufacturing Sdn Bhd, Rantau Plant).
2	Production Type	`MFGMOD_0`	Select **Standard** (Stock-based finished goods production).
3	Released Product	`ITMREF_0`	Enter product code (e.g. `FG-425X24`). System loads BOM 1 and Routing 1.
4	Released Quantity	`EXTQTY_0`	Enter target carton volume (e.g. `2,000` CTN = 48,000 bottles).
5	Start & End Dates	`STRDAT_0 / ENDDAT_0`	Define planned production start and completion shift dates and times.
6	Order Status	`MFGSTA_0`	Status 1 (Firm) confirms order for line dispatch; Status 2 (Planned) for scheduling.

3.2 Component Stock Reservation & Shortage Handling Engine

Clicking the **Allocation** button triggers the Sage X3 allocation engine across table `MFGMAT`:

- **Detailed Allocation:** Physical inventory lots (preforms, caps, labels, cartons) are locked exclusively for this Work Order. Other orders cannot consume these reserved lots.
- **Global Allocation:** Stock quantity is reserved at site level without specifying individual lot numbers.
- **Shortage Detection (`SHTQTY_0 > 0`):** If raw material stock is insufficient, Sage X3 displays a "*Material Shortage*" alert. Supervisors must contact Stores/Procurement to expedite Goods Receipts (`GESPTH`) in ERPDIS before releasing the line.

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4.0 SHOP FLOOR EXECUTION & PRODUCTION TRACKING

Actual shop floor output, scrap quantities, machine runtime, and material backflushing are recorded via `GESMTK`.

FLOWCHART 2: PRODUCTION OUTPUT RECORDING & MATERIAL BACKFLUSH ENGINE

STEP 1
Select Work Order
Select active WO from Left Drawer selection list.

→

STEP 2
Record Output/Scrap
Enter completed cartons and reject count.

→

STEP 3
Auto Backflushing
System deducts preforms, caps, labels, cartons.

→

STEP 4
Receipt to Stock Q
Finished goods enter quarantine status.

4.1 Operational Protocol for Production Tracking (GESMTK)

FIELD / ACTION	DATABASE FIELD (`MFGITMTRK`)	OPERATIONAL PROCEDURE & SYSTEM VALIDATION RULES
Tracking Type	`TRKTYP_0`	Select **Complete** for final shift run; select **Partial** if production continues next shift.
Produced Quantity	`CPLQTY_0`	Input physical count of good cartons packed onto pallets (e.g. `1,960` CTN).
Rejected Quantity	`REJCPLQTY_0`	Enter defective cartons/bottles (e.g. `40` CTN equivalent) for scrap tracking.
Batch / Lot Number	`LOT_0`	System generates lot number following BubblesO2 standard: `B{YYMMDD}-{Shift}`.
Stock Status	`STA_0`	**MANDATORY: STATUS Q (QUARANTINE)**. Field is system-locked to prevent manual release.
Machine Hours	`CPLWST_0` (in `MFGOPETRK`)	Actual machine run hours for each work center (e.g. `3.92` hours on `WST-FILL01`).
Labor Hours	`CPLLAB_0` (in `MFGOPETRK`)	Direct operator labor hours logged for shift personnel and line technicians.

4.2 Real-Time Material Backflushing & Scrap Reason Tracking

Upon clicking **Create**, Sage X3 executes two synchronized real-time transactions:

- **Automatic Component Backflushing (`MFGMATTRK`):** The system multiplies finished carton output by BOM ratios, instantly deducting 47,040 preforms, 47,040 caps, 47,040 labels, and 1,960 carton boxes from warehouse inventory.
- **Line Scrap Reason Categorization:** Rejects must be tagged with root-cause scrap codes (`SCR-BLOW` for wall deformity, `SCR-SEAL` for cap torque leak, `SCR-LBL` for torn shrink label, `SCR-BOX` for crushed carton).

5.0 QUALITY CONTROL (QC/QA) ARCHITECTURE & STOCK QUALITY ENGINE

The Quality Control (QC/QA) framework protects consumer safety and brand integrity by enforcing strict laboratory testing before bottled water can be distributed commercially.

5.1 Three-Tier Stock Quality Status Framework

STATUS Q (QUARANTINE)
Default initial state for 100% of finished goods upon production tracking. **System strictly locks this stock!** Sales Orders and Delivery Orders cannot allocate or dispatch Status Q lots.

STATUS A (ACCEPTED)
Granted only after physical, chemical, and microbiological lab tests pass all standards. Stock is released into commercial inventory, ready for customer sales allocation and shipping.

STATUS R (REJECTED)
Assigned to lots failing food safety limits (e.g. bacterial contamination or bad seal). Stock is permanently blocked and transferred to red reject quarantine for disposal or decanting.

5.2 Statutory Compliance & Food Safety Framework

CERTIFICATION / REGULATORY STANDARD	GOVERNING AUTHORITY	MANDATORY COMPLIANCE REQUIREMENT AT BUBBLESO2
Natural Mineral Water License	Ministry of Health Malaysia (KKM)	Food Regulations 1985, Regulation 360B governing underground source hygiene, bottling filtration, and daily chemical analysis.
HACCP & GMP Certification	SIRIM / KKM (MS 1480 / MS 1514)	Hazard Analysis Critical Control Points at borehole water extraction, 0.2µ microfiltration, and cleanroom bottle capping.
Halal Certification	JAKIM (MS 1500:2019)	Ensuring all packaging materials, lubricants, and processing aids are fully Halal compliant and free from contamination.
Food Safety Management System	ISO 22000 / FSSC 22000	End-to-end batch traceability from raw preform resin lot to finished carton delivery.

STRICT WAREHOUSE DISPATCH LOCKOUT
The ERPDIS distribution engine enforces an absolute constraint: If a lot number carries Status Q or Status R, the Delivery Order validation button (**GESSDH**) is completely disabled. Warehouse personnel cannot bypass this lock without authorized QA release in **GESSCS** .

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6.0 QC TECHNICAL INSPECTION SHEETS & TEST SPECIFICATIONS

Quality inspection parameters are formally standardized in Technical Control Sheets (`QLYCRD`) and Question Tables (`QLYCRDQST`).

6.1 BubblesO2 Technical Test Parameter Specifications (GESQCR / GESQST)

Every production batch of natural mineral water must undergo comprehensive testing recorded in function `GESQCR` :

TEST CODE	PARAMETER NAME	STANDARD LIMIT / TARGET	TESTING METHODOLOGY	REGULATORY STANDARD
`TEST-PH`	pH Balance	**7.20 – 7.80 pH**	Calibrated Digital pH Probe	Food Regulations 1985 (Reg 360B)
`TEST-TDS`	Total Dissolved Solids	**120 – 180 ppm (mg/L)**	Conductivity TDS Meter	Natural Mineral Water Standard
`TEST-TURB`	Turbidity	**< 0.50 NTU**	Digital Turbidimeter	Clarity & Optical Inspection
`TEST-VOL`	Net Fill Volume	**425.0 ml ± 5.0 ml**	Gravimetric Density Scale	Weights & Measures Act
`TEST-TORQ`	Cap Removal Torque	**1.50 – 2.20 Nm**	Digital Cap Torque Gauge	Capping Seal Integrity
`TEST-DROP`	Bottle Drop Resistance	**1.2m Drop (No Rupture)**	Vertical Impact Drop Rig	Packaging Transport Durability
`TEST-COLI`	Total Coliform Bacteria	**0 CFU / 100 ml (Zero)**	Membrane Filtration (37°C, 24h)	KKM Mandatory Pathogen Limit
`TEST-ECOL`	Escherichia coli (E. coli)	**0 CFU / 100 ml (Zero)**	Membrane Filtration (44.5°C, 24h)	KKM Mandatory Pathogen Limit
`TEST-PSEU`	Pseudomonas aeruginosa	**0 CFU / 250 ml (Zero)**	Selective Agar (37°C, 48h)	Bottled Water Microbiological Limit
`TEST-TVPC`	Total Viable Plate Count	**< 100 CFU / ml**	Nutrient Agar Pour Plate (37°C)	Microbiological Baseline Count

6.2 Recording Test Results in QLYCRDASW

When laboratory technicians execute testing in function `GESQCR` :

- **Numeric Measurements (`NUMASW_0`):** Quantitative values are logged (e.g. `pH = 7.45` , `TDS = 148.0` , `Torque = 1.85`).
- **Microbiological Observations (`ALPASW_0`):** Pathogen tests must read "*NOT DETECTED*" or "*0 CFU*".
- **Automated Pass/Fail Flag (`FLGASW_0`):** If all measured values fall within the pre-set tolerance boundaries, Sage X3 sets `FLGASW_0 = 1` (Pass). Any deviation automatically flags `FLGASW_0 = 2` (Non-Conformance).

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7.0 STATISTICAL SAMPLING PLANS & LABORATORY INSPECTION

To ensure statistical quality confidence without destroying excessive stock, sampling protocols adhere to **ISO 2859-1 (Sampling Procedures for Inspection by Attributes)**.

FLOWCHART 3: INTEGRATED QUALITY INSPECTION, SAMPLING & LAB RELEASE WORKFLOW

STEP 1
Production Receipt
Finished goods enter Status Q via GESMTK.

→

STEP 2
Generate Sampling
GESSMP calculates ISO sample size.

→

STEP 3
48-Hr Incubation
Physical sample culture in microbiology lab.

→

STEP 4
Quality Decision
Evaluate results & approve stock release.

7.1 ISO 2859 Sampling Plan Table (STOQLYSMP / CODSMP_0)

BATCH LOT SIZE (CARTONS)	INSPECTION LEVEL	SAMPLE SIZE (DRAWN)	ACCEPTANCE (AC)	REJECTION (RE)
501 – 1,200 Cartons	General Level II	**80 Bottles**	**0 Defect**	**1 Defect**
1,201 – 3,200 Cartons	General Level II	**125 Bottles**	**0 Defect**	**1 Defect**
3,201 – 10,000 Cartons	General Level II	**200 Bottles**	**0 Defect**	**1 Defect**

7.2 Mandatory 48-Hour Microbiological Incubation Holding Period

In strict accordance with Ministry of Health (KKM) regulations for natural mineral water:

- **Culture Incubation Hold:** Microbiological testing requires incubation (24 hours for Coliforms at 37°C, 48 hours for *Pseudomonas aeruginosa*).
- **Time-Lock Rule:** No lot of bottled mineral water may be converted from Status Q to Status A until exactly **48 hours have elapsed** from the completion timestamp of production tracking.
- **Retained Reference Samples:** An additional 12 bottles per batch are preserved in the QA Retention Archive for 24 months (full product shelf-life) for post-market auditing.

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8.0 QUALITY DISPOSITION, NON-CONFORMANCE & STOCK RELEASE

Following the completion of laboratory tests, the Quality Assurance Executive executes the formal stock status change in `GESSCS`.

8.1 Quality Disposition Decision Matrix

INSPECTION OUTCOME	FLAG (`FLGASW`)	QUALITY ACTION & PROTOCOL	WAREHOUSE INVENTORY IMPACT
Full Compliance	`1` (All Pass)	**Approve for Commercial Release:** QA Head approves and generates Certificate of Analysis (COA).	Execute `GESSCS`: Change lot from **Status Q to Status A**. Released for sales allocation.
Minor Deviation	`2` (Borderline)	**Quarantine Alert & Retesting:** Issue Non-Conformance Alert (`QLYCRDNQA`) and draw secondary sample.	Lot remains in **Status Q**. Warehouse dispatch remains locked pending QA manager review.
Critical Failure	`2` (Pathogen)	**Immediate Lot Rejection:** Complete quarantine, physical red tagging, and disposal order.	Execute `GESSCS`: Change lot from **Status Q to Status R**. Move to scrap area for bottle decanting.

8.2 Executing Stock Status Conversion (GESSCS — Stock Change)

To release an approved batch into commercial stock:

1. Open function `GESSCS` (*Stock* > *Internal movements* > *Stock change*).
2. Confirm Stock Site = `BOMSB` (Finished Goods Warehouse).
3. Enter Product Code (e.g. `FG-425X24`) and select the Status Q Lot in the Left Selection Drawer.
4. Change **New Status to** `A` (or `R` if rejected).
5. Enter the Certificate of Analysis reference (e.g. `COA-2609-0425`) and reason code.
6. Click **Save**. The lot is instantly visible in `ERPDIS` for customer sales order allocation and delivery.

8.3 Digital Certificate of Analysis (COA) Issuance

Major commercial retail accounts (Lotus's, TF Value-Mart, Shell, Petronas) require an official COA accompanying every delivery consignment. Sage X3 automatically compiles this document directly from recorded values in `QLYCRDASW`, displaying verified pH, TDS, turbidity, and zero-pathogen clearance.

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9.0 MANUFACTURING COSTING, WIP VALUATION & ORDER CLOSING

Manufacturing accounting reconciles shop floor consumption against financial inventory valuations, transferring Work-in-Progress (WIP) into finished goods assets.

FLOWCHART 4: PRODUCTION COST CALCULATION, WIP SETTLEMENT & ORDER CLOSING WORKFLOW

STEP 1
Confirm Outputs
Verify all MTK tracking tickets recorded.

→

STEP 2
Calculate Costs
Run CSTMFG cost rollup calculation.

→

STEP 3
Variance Analysis
Analyze Material, Labor & Machine variances.

→

STEP 4
Close Work Order
Execute FUNMCLO & settle WIP to GL.

9.1 Four-Tier Production Cost Structure (MFGCOST)

COST COMPONENT	DATABASE FIELD (MFGCOST)	VALUATION METHODOLOGY & ACCOUNTING BASIS
Raw Materials Cost	`MATTOT_0`	Actual consumption of preforms, caps, labels, cartons, and water at moving average cost.
Machine Running Cost	`MACTOT_0`	Machine hourly cost center absorption rate multiplied by recorded runtime hours.
Direct Labor Cost	`LABTOT_0`	Standard operator hourly wage rate applied to logged operation and setup hours.
Plant Overheads	`SCOTOT_0`	Factory utilities, cleanroom HEPA maintenance, laboratory QA testing, and depreciation.

9.2 Production Variance Analysis & Work Order Closure (FUNMCLO)

Before an order is administratively closed, cost accountants evaluate production variances:

- **Material Usage Variance:** Difference between standard BOM quantities and actual consumed components (reflects line blow loss or preform wastage).
- **Operational Efficiency Variance:** Difference between standard routing machine cycle hours and actual recorded hours.
- **Work Order Closure (`FUNMCLO`):** Sets order status to *Closed*, posts the auto-journal `WOO` clearing Account 2100 (WIP) into Account 2200 (Finished Goods), and recognizes cost variances into Account 7110.

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10.0 AUTOMATIC ACCOUNTING ENGINE & DATABASE SCHEMA LINKAGE

Shop floor events automatically trigger financial accounting entries via the Automatic Accounting Engine (`GESGAU`).

10.1 Operational Event to Auto-Journal Mapping

OPERATIONAL PRODUCTION EVENT	FUNCTION	AUTO-JOURNAL (GAU)	FINANCIAL ACCOUNTING DEBIT / CREDIT IMPACT
Material Consumption	`GESMTK`	`WOO` (Work Order)	**Debit:** Work-in-Progress (2100) **Credit:** Raw Materials Inventory (2000)
Finished Goods Receipt	`GESMTK`	`WOO` (Work Order)	**Debit:** Finished Goods Inventory (2200) **Credit:** Work-in-Progress (2100)
Work Center Machine/Labor	`GESMTK`	`WOO` (Work Order)	**Debit:** Work-in-Progress (2100) **Credit:** Factory Overhead Applied (7100)
Quality Rejection (Scrap)	`GESSCS`	`STK` (Stock Change)	**Debit:** Production Scrap Expense (7120) **Credit:** Finished Goods Inventory (2200)

10.2 Deep Database Schema Architecture from Database.sql

Extracted directly from `Database.sql` (8,229 tables, schema `[BASE]`), the core manufacturing and quality database architecture includes:

SQL TABLE NAME	PRIMARY KEY & KEY FIELDS	TABLE ROLE & BUSINESS APPLICATION IN SAGE X3
`BOM` / `BOMD`	`ITMREF_0, BOMALT_0, BOMSEQ_0`	Master Bill of Materials header and component item lines, scrap factors (`SCA_0`), and operation links.
`ROUTING` / `ROUOPE`	`ITMREF_0, ROUALT_0, OPENUM_0`	Manufacturing routings, work centers (`WST_0`), machine run times, and Quality Sheet links (`TECCRD_0`).
`MFGHEAD`	`MFGNUM_0, MFGFCY_0, MFGSTA_0`	Master Work Order header storing target quantity (`EXTQTY_0`), completed quantity, and schedule dates.
`MFGMAT`	`MFGNUM_0, MFGLIN_0, ITMREF_0`	Work Order component allocation table tracking required, allocated, and shortage quantities (`SHTQTY_0`).
`MFGOPE`	`MFGNUM_0, OPENUM_0, WST_0`	Work Order operation progress recording planned vs actual machine hours (`CPLWST_0`) and labor hours.
`QLYCRD` / `QLYCRDQST`	`QLYCRD_0, QSTNUM_0, NUMLIN_0`	Technical Quality Sheets and parameter questions (pH, TDS, turbidity, torque, microbiological limits).
`QLYCRDASW`	`QLYCTLDEM_0, QLYCRD_0, QSTNUM_0`	Quality test recording storing numeric results (`NUMASW_0`), textual answers, and pass/fail flags (`FLGASW_0`).
`STOQLYSMP`	`VCRNUM_0, ITMREF_0, LOT_0`	Statistical sampling records tracking drawn sample size (`QTYSMP_0`), accepted units, and sampling status.
`MFGCOST`	`VCRNUM_0, ITMREF_0, UID_0`	Work Order cost rollup tracking material (`MATTOT_0`), machine (`MACTOT_0`), and labor totals.

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11.0 TROUBLESHOOTING MATRIX & AUTHORITATIVE ENTERPRISE GLOSSARY

11.1 Troubleshooting Matrix for Common Manufacturing & QC Exceptions

EXCEPTION MESSAGE / SYMPTOM	ROOT SYSTEM CAUSE	STANDARD CORRECTIVE ACTION
"Component shortage / Material not allocated"	Warehouse stock of preforms, caps, labels, or boxes is below required WO volume.	Open `CONSSST` to identify short items. Request Stores to expedite GRN receipt in ERPDIS (`GESPTH`).
"Stock lot locked in Status Q — Cannot deliver"	Finished bottled water lot has not been released from QC quarantine.	Check lab results in `GESQCR` . If all micro/chemical tests pass, execute status change to **A** in `GESSCS` .
"Parameter value outside specification limits"	Recorded test reading (e.g. pH = 8.10 or cap torque = 1.1 Nm) breaches quality thresholds.	System flags non-conformance (`FLGASW_0 = 2`). Initiate secondary sampling. If persistent, move lot to Status R.
"Work Order cannot be closed — Open operations"	Unrecorded machine operations or unfinished tracking tickets in `GESMTK` .	Review operation progress in `GESMTK` . Complete remaining operation entries or adjust remaining target quantity to zero.
"High material variance detected during costing"	Excessive line breakage of preforms or damaged cartons during packaging.	Review scrap logs in `GESMTK` . If scrap is chronic, submit engineering request to update BOM scrap factor (`SCA_0`).

11.2 Authoritative Manufacturing & Quality Control Glossary

• **Bill of Materials (BOM):** The authoritative recipe defining the exact quantities of raw materials needed for one unit of finished goods.

• **Manufacturing Routing:** The sequential path of work centers and machine operations through which raw materials are transformed.

• **Work Order (WO):** Formal enterprise authorization detailing product code, target quantity, component reservations, and schedule.

• **Component Backflushing:** Automatic deduction of raw material stock upon recording finished good output based on BOM ratios.

• **Status Q (Quarantine):** Initial mandatory quarantine state locking finished goods from sales allocation pending lab clearance.

• **Status A (Accepted):** Verified commercial release status confirming compliance with food safety and mineral water standards.

• **Status R (Rejected):** Permanently locked non-conforming status for scrap, decanting, bottle recycling, or disposal.

• **AQL / ISO 2859 Sampling:** Statistical attribute sampling defining the exact number of sample units drawn per lot size.

• **Certificate of Analysis (COA):** Formal quality compliance document certifying physical, chemical, and microbiological test results.

• **Work-in-Progress (WIP):** Asset account (Account 2100) aggregating costs of materials, labor, and machine overhead during production.

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